

Do circulating leucocytes and lymphocyte subtypes increase in response to brief exercise in children with and without asthma?

Background Exercise can alter health in children in both beneficial (eg reduced long-term risk of atherosclerosis) and adverse (eg exercise-induced asthma) ways. The mechanisms linking exercise and health are not known, but may rest, partly, on the ability of exercise to increase circulating immune cells. Little is known about the effect of brief exercise, more reflective of naturally occurring patterns of physical activity in children, on immune cell responses. **Objectives** To determine whether (1) a 6-min bout of exercise can increase circulating inflammatory cells in healthy children and (2) the effect of brief exercise is greater in children with a history of asthma. **Methods** Children with mild-moderate persistent asthma and age-matched controls (n = 14 in each group, mean age 13.6 years) performed a 6-min bout of cycle-ergometer exercise. Spirometry was performed at baseline and after exercise. Blood was drawn before and after exercise, leucocytes were quantified and key lymphocyte cell surface markers were assessed by flow cytometry. **Results** Exercise decreased spirometry only in children with asthma, but increased ($p < 0.001$) most types of leucocytes (eg lymphocytes (controls, mean (SD) 1210 (208) cells/ μ l; children with asthma, 1119 (147) cells/ μ l) and eosinophils (controls, 104 (22) cells/ μ l; children with asthma, 88 (20) cells/ μ l)) to the same degree in both groups. Similarly, exercise increased T helper cells (controls, 248 (60) cells/ μ l; children with asthma, 232 (53) cells/ μ l) and most other lymphocyte subtypes tested. By contrast, although basophils (16 (5) cells/ μ l) and CD4⁺ CD45RO⁺ RA⁺ lymphocytes (19 (4) cells/ μ l) increased in controls, no increase in these cell types was found in children with asthma. **Conclusions** Exercise increased many circulating inflammatory cells in both children with asthma and controls. Circulating inflammatory cells did increase in children with asthma, but not to a greater degree than in controls. In fact, basophils and T helper lymphocyte memory transition cells did not increase in children with asthma, whereas they did increase in controls. Even brief exercise in children and adolescents robustly mobilises circulating immune cells.